

Non Right Angled Triangles

Solutions



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ACMMG: 273

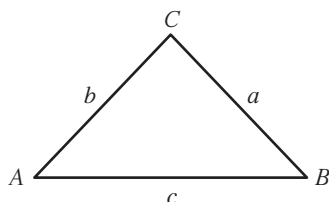


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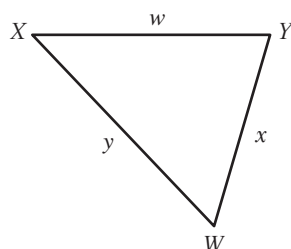
Page 3 questions

1. Label the sides in the following triangles.

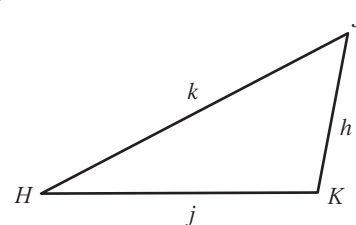
a



b

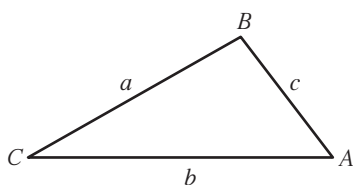


c

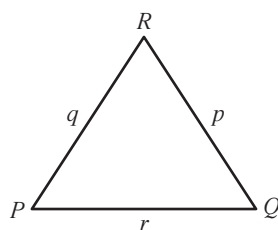


2. Label the angles in the following triangles.

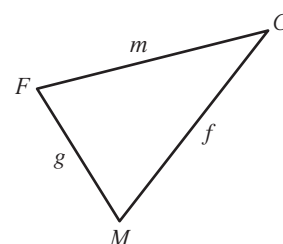
a



b

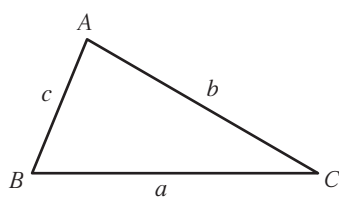


c

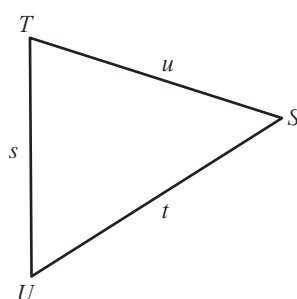


3. Complete the labelling of the following triangles.

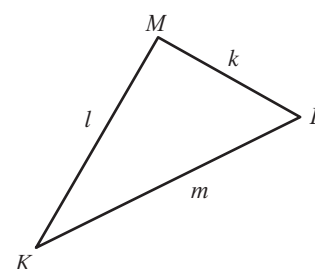
a



b



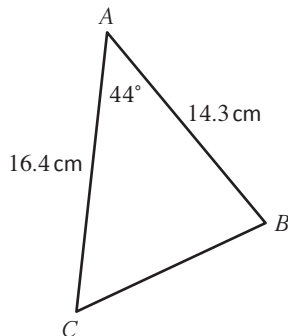
c



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1. Find the area of the following triangles to 1 decimal place.

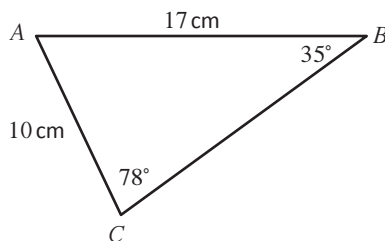
a



b and c are given, so we will use $\text{Area} = \frac{1}{2} bc \sin A$

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} bc \sin A \\ &= \frac{1}{2}(16.4)(14.3)\sin 44^\circ \\ &= 81.456... \approx 81.5\text{cm}^2\end{aligned}$$

b



The interior angle between the two sides is not given so calculate the angle A

$$\begin{aligned}\angle A + 78^\circ + 35^\circ &= 180^\circ \\ \therefore \angle A &= 67^\circ\end{aligned}$$

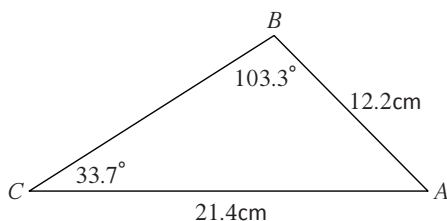
b and c are given, so we will use $\text{Area} = \frac{1}{2} bc \sin A$

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} bc \sin A \\ &= \frac{1}{2}(10)(17)\sin 67^\circ \\ &= 78.2429... \approx 74.2\text{cm}^2\end{aligned}$$

2. $\triangle ABC$ has $\angle B = 103.3^\circ$, $\angle C = 33.7^\circ$, $b = 21.4\text{cm}$ and $c = 12.2\text{cm}$.

a

Draw the triangle.



b

Find $\angle A$ to the nearest degree.

$$\begin{aligned}\angle A + 103.3^\circ + 33.7^\circ &= 180^\circ \\ \therefore \angle A &= 43^\circ\end{aligned}$$

c

Find the area of the triangle to 1 decimal place.

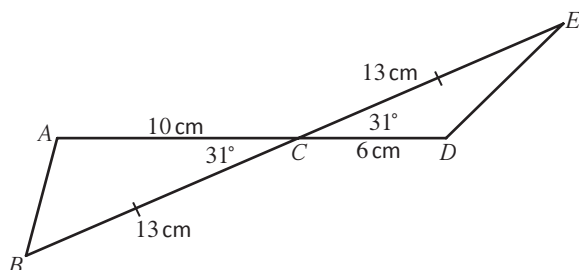
b and c are given, so we will use $\text{Area} = \frac{1}{2} bc \sin A$

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} bc \sin A \\ &= \frac{1}{2}(21.4)(12.2)\sin 43^\circ \\ &= 89.028... \approx 89.0\text{cm}^2\end{aligned}$$

Page 6 questions

3. Can you find the areas of the following shapes to the nearest square cm?

- a This can be labelled and redrawn like so:



The lines BC and CE are shown as equal by the marks across each line.

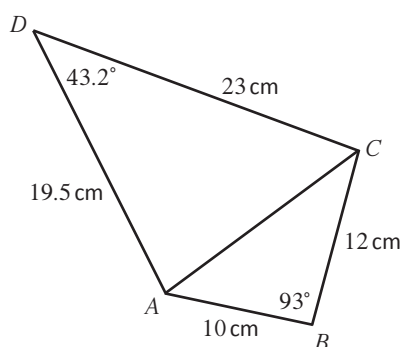
The angles at C in each triangle are equal.

$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2}(13)(10)\sin 31^\circ \\ &= 33.477\dots\end{aligned}$$

$$\begin{aligned}\text{Area } \triangle CDE &= \frac{1}{2} bc \sin C \\ &= \frac{1}{2}(13)(6)\sin 31^\circ \\ &= 20.086\dots\end{aligned}$$

$$\begin{aligned}\therefore \text{Total area} &= 33.477\dots + 20.086\dots \\ &= 53.563\dots \approx 54\text{cm}^2\end{aligned}$$

- b Hint: Draw in Diagonal AC



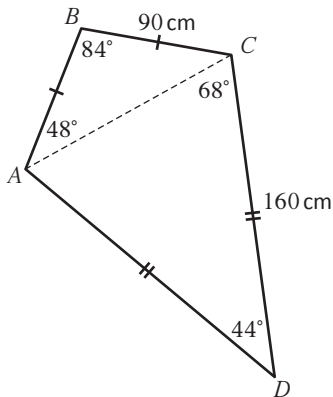
$$\begin{aligned}\text{Area} = \triangle ABC &= \frac{1}{2} (BC)(AB) \sin 93^\circ \\ &= \frac{1}{2}(12)(10) \sin 93^\circ \\ &= 59.91777\dots\end{aligned}$$

$$\begin{aligned}\text{Area} = \triangle ACD &= \frac{1}{2} (AD)(CD) \sin 43.2^\circ \\ &= \frac{1}{2}(19.5)(23) \sin 43.2^\circ \\ &= 153.5096\dots\end{aligned}$$

$$\begin{aligned}\therefore \text{Total area} &= 59.9177\dots + 153.5096\dots \\ &= 213.4273\dots \approx 213\text{cm}^2\end{aligned}$$

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4. Luke wants to make a kite in the shape below. How many square cm of material will he need?



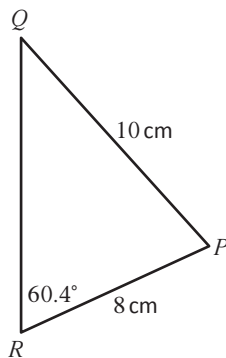
$$\begin{aligned}\text{Area} = \triangle ABC &= \frac{1}{2}(BC)^2 \sin 84^\circ \\ &= \frac{1}{2}(90)^2 \sin 84^\circ \\ &= 4027.8136\dots\end{aligned}$$

$$\begin{aligned}\text{Area} = \triangle ACD &= \frac{1}{2}(CD)^2 \sin 44^\circ \\ &= \frac{1}{2}(160)^2 \sin 44^\circ \\ &= 8891.627\dots\end{aligned}$$

$$\therefore \text{Total area} = 12919.44\dots \approx 12919\text{cm}^2$$

Both of these triangles are isosceles triangles, ie they have two sides the same length each.

5. The area of the triangle $\triangle PQR$ is 38.6 cm^2 . Find $\angle Q$. (Hint: Find $\angle P$ first)

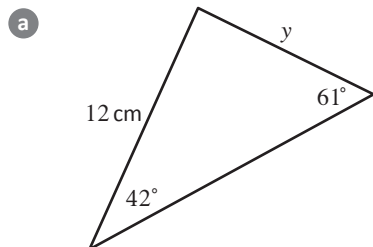


$$\begin{aligned}\text{Area } \triangle PQR &= \frac{1}{2}qr \sin P \\ 38.6 &= \frac{1}{2}(8)(10)\sin P \\ \therefore \sin P &= 0.965 \\ \therefore P &= 74.8^\circ\end{aligned}$$

$$\begin{aligned}\angle Q + 74.8^\circ + 60.4^\circ &= 180^\circ \\ \therefore \angle Q &= 44.8^\circ\end{aligned}$$

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The Sine Rule

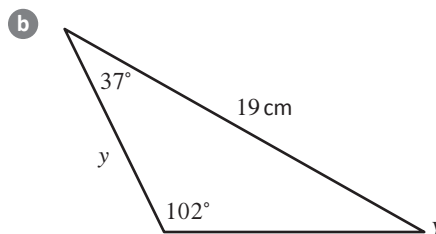
1. Find the length y in the triangles below to 1 decimal place.

Apply the sine rule:

$$\frac{\sin 42^\circ}{y} = \frac{\sin 61^\circ}{12}$$

$$y = \frac{12 \times \sin 42^\circ}{\sin 61^\circ}$$

$$y = 9.2 \text{ cm}$$



$$\angle Y + 102^\circ + 37^\circ = 180^\circ$$

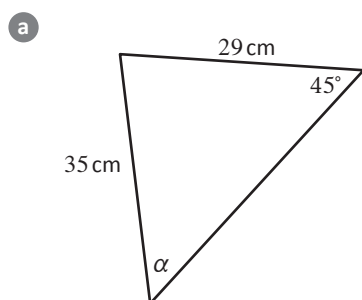
$$\therefore \angle Y = 41^\circ$$

Apply the sine rule:

$$\frac{\sin 41^\circ}{y} = \frac{\sin 102^\circ}{19}$$

$$y = \frac{19 \times \sin 41^\circ}{\sin 102^\circ}$$

$$y = 12.7 \text{ cm}$$

2. Find the size of angle α in the following triangles:

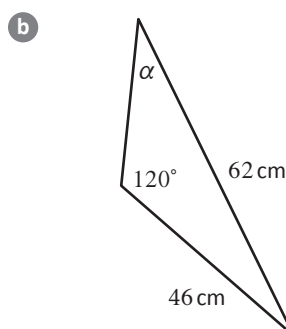
Apply the sine rule:

$$\frac{\sin \alpha}{29} = \frac{\sin 45^\circ}{35}$$

$$\therefore \sin \alpha = 29 \times \frac{\sin 45^\circ}{35}$$

$$\sin \alpha = 0.586$$

$$\therefore \alpha = 35.9^\circ$$



Apply the sine rule:

$$\frac{\sin \alpha}{46} = \frac{\sin 120^\circ}{62}$$

$$\sin \alpha = 46 \times \frac{\sin 120^\circ}{62}$$

$$\sin \alpha = 0.643$$

$$\therefore \alpha = 40^\circ$$

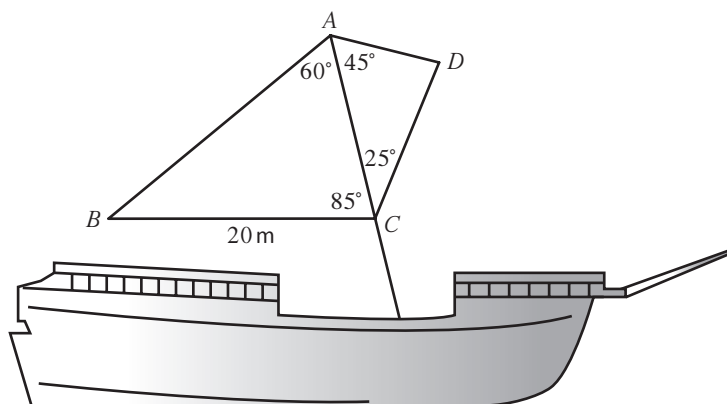
Page 11 questions

3. A ship has sails as drawn in the diagram below.

- a Find the length of AC .

$$\begin{aligned}\angle ABC &= 180^\circ - 60^\circ - 85^\circ \\ &= 35^\circ\end{aligned}$$

$$\begin{aligned}\therefore \frac{AC}{\sin B} &= \frac{20}{\sin 60^\circ} \\ AC &= \frac{20 \sin 35^\circ}{\sin 60^\circ} \\ &= 13.2461804\dots\text{m} \\ &\approx 13.2\text{m (to 1 d.p.)}\end{aligned}$$



- b Find the lengths of DC and AD .

$$\begin{aligned}\angle ADC &= 180^\circ - 45^\circ - 25^\circ \\ &= 110^\circ\end{aligned}$$

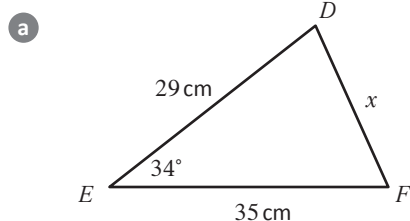
$$\begin{aligned}\therefore \frac{DC}{\sin 45^\circ} &= \frac{13.2461804}{\sin 110^\circ} \\ DC &= \frac{13.2461804 \sin 45^\circ}{\sin 110^\circ} \\ &= 9.96758278\dots\text{m} \\ &\approx 10.0\text{m (to 1 d.p.)}\end{aligned}$$

$$\begin{aligned}\therefore \frac{AD}{\sin 25^\circ} &= \frac{13.2461804}{\sin 110^\circ} \\ AD &= \frac{13.2461804 \times \sin 25^\circ}{\sin 110^\circ} \\ &= 5.957484817\dots\text{m} \\ &\approx 6.0\text{m (to 1 d.p.)}\end{aligned}$$

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The Cosine Rule

4. Find the length of the missing side x in each of the following to 1 decimal place.



Apply the cosine rule

$$e^2 = d^2 + f^2 - 2df \cos E$$

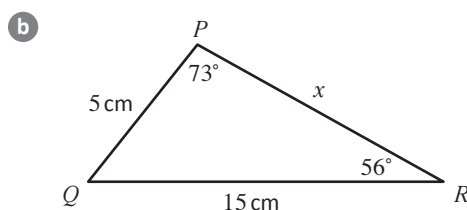
In this case:

$$x^2 = 29^2 + 35^2 - (2 \times 29 \times 35) \cos 34^\circ$$

$$x^2 = 2066 - (2030) \cos 34^\circ$$

$$x = \sqrt{383.05}$$

$$\therefore x = 19.6 \text{ cm (to 1 d.p.)}$$



Find Q

$$Q + 73^\circ + 56^\circ = 180^\circ$$

$$\therefore Q = 51^\circ$$

In this case

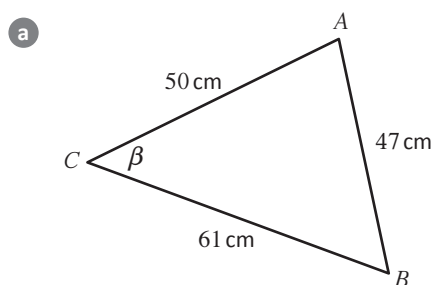
$$x^2 = p^2 + r^2 - 2pr \cos Q$$

$$x^2 = 15^2 + 5^2 - (2 \times 15 \times 5) \cos 51^\circ$$

$$x = \sqrt{155.6019...}$$

$$\therefore x = 12.5 \text{ cm (to 1 d.p.)}$$

5. Find the value of the angle β in each of the following to the nearest degree.



Apply the cosine rule

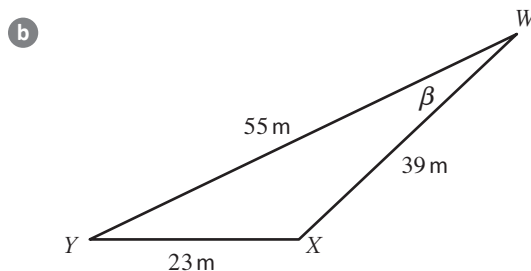
$$c^2 = a^2 + b^2 - 2ab \cos \beta$$

$$47^2 = 50^2 + 61^2 - 2 \times 50 \times 61 \cos \beta$$

$$\beta = \cos^{-1} \left(\frac{47^2 - 50^2 - 61^2}{-2 \times 50 \times 61} \right)$$

$$\therefore \beta = 48.8749...$$

$$\therefore \beta \approx 49^\circ \text{ to nearest degree}$$



Apply the cosine rule

$$w^2 = x^2 + y^2 - 2xy \cos \beta$$

$$23^2 = 55^2 + 39^2 - 2 \times 55 \times 39 \cos \beta$$

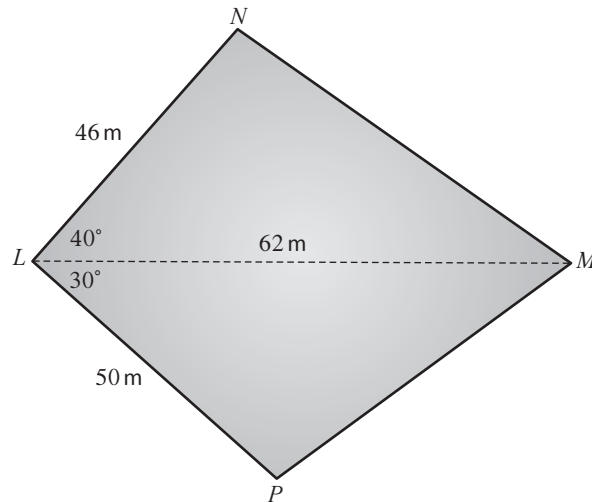
$$\beta = \cos^{-1} \left(\frac{23^2 - 55^2 - 39^2}{-2 \times 55 \times 39} \right)$$

$$\therefore \beta = 20.5504...$$

$$\therefore \beta \approx 21^\circ \text{ to nearest degree}$$

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6. A canyon has the following dimensions:



To walk around this canyon, would it be shorter to walk along LN or LPM ?

Find the distance NM

$$NM^2 = LM^2 + LN^2 - 2 \times LM \times LN \cos 40^\circ$$

$$NM = \sqrt{62^2 + 46^2 - 2 \times 62 \times 46 \cos 40^\circ}$$

$$NM = 39.9\text{m (to 1 d.p.)}$$

$$\therefore LNM = 46\text{m} + 39.9\text{m} = 85.9\text{m}$$

Find the distance PM

$$PM^2 = LM^2 + LP^2 - 2 \times LM \times LP \cos 30^\circ$$

$$PM = \sqrt{62^2 + 50^2 - 2 \times 62 \times 50 \cos 30^\circ}$$

$$PM = 31.2\text{m (to 1 d.p.)}$$

$$\therefore LPM = 50\text{m} + 31.2\text{m} = 81.2\text{m}$$

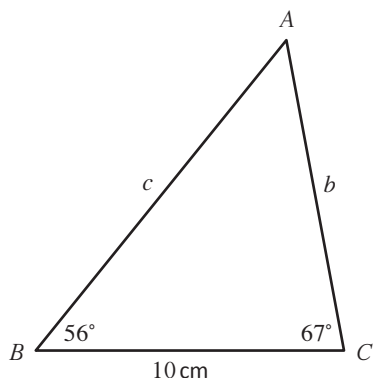
$\therefore$ it would be shorter to walk along LPM than LN

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Both Rules

7. Use the sine and cosine rules to find the missing sides and angles in these triangles to 2 decimal places.

a



$$\angle A = 180^\circ - (56^\circ + 67^\circ) = 57^\circ$$

Apply the sine rule to find b

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 57^\circ}{10} = \frac{\sin 56^\circ}{b}$$

$$b = \frac{10 \sin 56^\circ}{\sin 57^\circ}$$

$$\therefore b = 9.885139699$$

$$\therefore b \approx 9.89 \text{ cm (to 2 d.p.)}$$

Apply the sine rule to find c

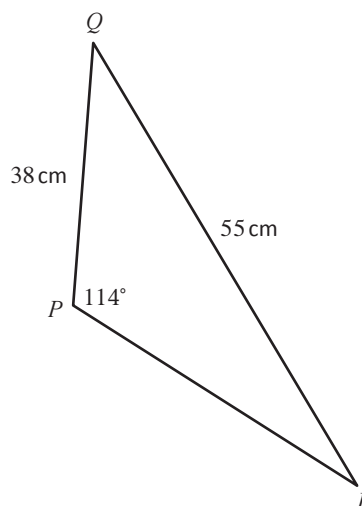
$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 10^2 + 9.89^2 - 2 \times 10 \times 9.89 \cos 67^\circ$$

$$\therefore c = 10.9784...$$

$$\therefore c \approx 10.99 \text{ cm (to 2 d.p.)}$$

b



Apply the sine rule to find $\angle R$

$$\frac{\sin 114^\circ}{55} = \frac{\sin R}{38}$$

$$\therefore \angle R = \sin^{-1} \left(38 \times \frac{\sin 114^\circ}{55} \right)$$

$$\therefore \angle R = 39.137^\circ$$

$$\approx 39.14^\circ \text{ (to 2 d.p.)}$$

$$\therefore \angle Q = 180^\circ - (114^\circ + 39.14^\circ)$$

$$= 26.86^\circ \text{ (to 2 d.p.)}$$

Apply the cosine rule to find q

$$q^2 = p^2 + r^2 - 2pr \cos Q$$

$$\therefore q^2 = 55^2 + 38^2 - 2 \times 55 \times 38 \cos 26.86^\circ$$

$$\therefore q = 27.20 \text{ cm (to 2 d.p.)}$$

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1. In triangle $\triangle ABC$ $a = 19$ cm, $b = 15$ cm and $\angle B = 20^\circ$.

- a Find two possible values for $\angle A$ to the nearest degree.

Apply the sine rule to find $\angle A$

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{19} = \frac{\sin 20^\circ}{15}$$

$$\therefore A = \sin^{-1} 0.4322\dots$$

$$\begin{aligned} \therefore \angle A &= 25.67^\circ & \text{or} & & \angle A &= (180^\circ - 26^\circ) \\ &= 26^\circ \text{ (to nearest degree)} & & & &= 154^\circ \end{aligned}$$

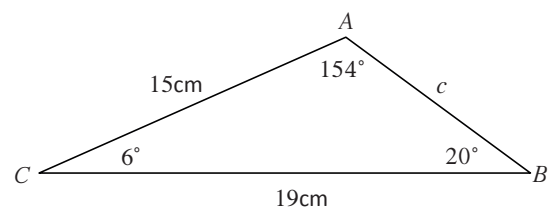
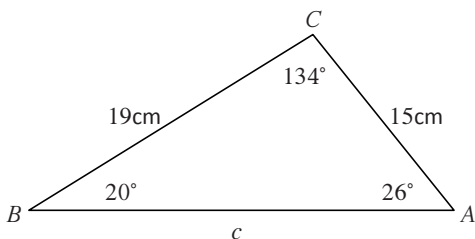
- b Check if these answers are valid. Is this an ambiguous case?

$$\begin{aligned} \therefore \angle A + \angle B &= 26^\circ + 20^\circ \\ &= 46^\circ < 180^\circ \quad \therefore \text{valid} \end{aligned}$$

$$\begin{aligned} \therefore \angle A + \angle B &= 154^\circ + 20^\circ \\ &= 174^\circ < 180^\circ \quad \therefore \text{valid} \end{aligned}$$

Both answers are valid; therefore this is an ambiguous case.

- c Draw a rough sketch of the triangle(s).



- d Find the third angle in the triangle(s) and add them to the diagram in c.

$$\angle C = 180^\circ - (\angle A + \angle B) \quad \text{or} \quad \angle C = 180^\circ - (154^\circ + 20^\circ)$$

$$\begin{aligned} \angle C &= 180^\circ - (26^\circ + 20^\circ) & & = 6^\circ \\ &= 134^\circ \end{aligned}$$

Page 20 questions

2. In $\triangle PQR$, $\angle R = 40^\circ$, $p = 20.3\text{cm}$ and $r = 13.9$.

- a Find the two possible values for $\angle P$ to the nearest degree.

Apply the sine rule to find $\angle P$

$$\frac{\sin P}{p} = \frac{\sin R}{r}$$

$$\frac{\sin P}{20.3} = \frac{\sin 40^\circ}{13.9}$$

$$\therefore P = \sin^{-1} 0.9387...$$

$$\begin{aligned} \therefore \angle P &= 69.84...^\circ & \text{or} & & \angle P &= (180^\circ - 70^\circ) \\ &\approx 70^\circ \text{ (to nearest degree)} & & & &= 110^\circ \end{aligned}$$

- b Is this an ambiguous case (are both angles valid)?

$$\begin{aligned} \angle P + \angle R &= 70^\circ + 40^\circ \\ &= 110^\circ < 180^\circ \quad \therefore \text{valid} \end{aligned}$$

$$\begin{aligned} \angle P + \angle R &= 110^\circ + 40^\circ \\ &= 150^\circ < 180^\circ \quad \therefore \text{valid} \end{aligned}$$

Both answers are valid; therefore this is an ambiguous case.

- c Which answer is valid if $\triangle PQR$ is acute? Which answer is invalid?

If $\triangle PQR$ is acute, the angles are in the range $0^\circ - 90^\circ$, $\angle P = 70^\circ$ is the valid answer and $\angle P = 110^\circ$ is invalid.

- d Which answer is valid if $\angle P$ is obtuse? Which answer is invalid?

If $\triangle PQR$ is obtuse, the angles are in the range $90^\circ - 180^\circ$, $\angle P = 110^\circ$ is the valid answer and $\angle P = 70^\circ$ is invalid.

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3. In triangle $\triangle JKL$, $\angle K = 70^\circ$, $k = 51.4\text{cm}$ and $l = 42\text{cm}$.

- a Find the valid size(s) of $\angle L$.

Apply the sine rule to find $\angle L$ (to the nearest degree)

$$\frac{\sin L}{l} = \frac{\sin K}{k}$$

$$\frac{\sin L}{42} = \frac{\sin 70^\circ}{51.4}$$

$$\therefore L = \sin^{-1} 0.7678\dots$$

$$\begin{aligned} \therefore \angle L &= 50.16 & \text{or} & \therefore \angle L = (180^\circ - 50^\circ) \\ &= 50^\circ \text{ (to nearest degree)} & & = 130^\circ \text{ (to nearest degree)} \end{aligned}$$

but $\angle L = 130^\circ$ is not a valid answer as:

$$\begin{aligned} \angle L + \angle K &= 130^\circ + 70^\circ \\ &= 200^\circ > 180^\circ \quad \therefore \text{invalid} \end{aligned}$$

$$\therefore \angle L = 50^\circ$$

- b Find $\angle J$.

$$\angle J = 180^\circ - (\angle L + \angle K)$$

$$\angle J = 180^\circ - (50^\circ + 70^\circ) = 60^\circ$$

- c Find the length of j .

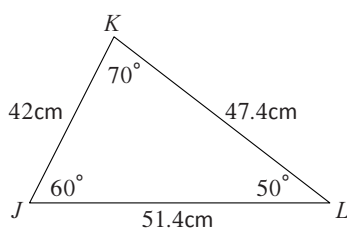
$$\frac{\sin J}{j} = \frac{\sin K}{k}$$

$$j = \frac{k \sin J}{\sin K}$$

$$\therefore j = 47.370\dots$$

$$\therefore j = 47.4\text{cm (to 1 d.p.)}$$

- d Draw $\triangle JKL$.





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